

Suppose you have a deterministic function $f(X)$ where X is an array of d Boolean values, and f outputs a Boolean value that is some function of those inputs. The function can be arbitrarily complicated, but for the same input x , it always outputs the same value $f(x)$.

Suppose we collect a *lot* of data sampled from this function, such that we see each of the possible 2^d values of x at least once. Assume our data collection doesn't have any noise.

Can a decision tree be built to perfectly represent $f(X)$? If so, give an explanation or simple proof of why such a decision tree exists. If not, provide an explanation or counterexample of a Boolean function that can't be represented by a decision tree.

Yes; decision tree functions are a subset of Boolean functions that can be represented by a tree-like structure. Each internal node would represent a test on one of the input variables.

Proof: There are 2^d possible input values for X , and that for each input value, the function $f(X)$ outputs a Boolean value. Since we have collected data for each possible input value and the function is deterministic, we can use this data to build a decision tree.